



Introduction to

# CLOUD COMPUTING

# Let's Start With Two Questions



How many of you use Google Drive, Gmail, YouTube, or WhatsApp?



Where are your photos, emails, and videos actually stored? Inside your mobile phone?

**Actually, No.** Most of your data is stored in huge computers called **servers**, located somewhere else in the world. You access it through the Internet.

***This concept is called Cloud Computing.***

# What Is Cloud Computing?

Cloud Computing means using computing services — **storage, software, databases, and servers** — through the Internet, instead of your own computer.

***In simple words:** Instead of buying and maintaining a powerful computer, you rent computing resources whenever you need them — through the Internet.*



Storage



Software



Servers



Databases

# First, Understand the Word "Cloud"

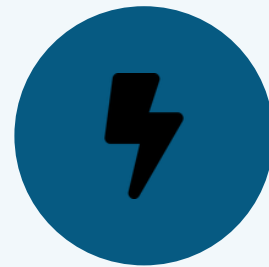


A collection of powerful  
computers (servers)  
connected via the Internet

*These servers live in massive buildings called Data Centers — and they run 24 hours a day, every day.*



**Thousands of  
Servers**



**Powered  
24 / 7**



**Guarded &  
Secured**



**Connected  
via  
Internet**

# Real-Life Example 1 — Google Drive

## The Scenario

**5 GB** of photos

**10 GB** of videos

*Your mobile has only 2 GB free space.*

**What do you do? Instead of deleting files, you upload them to Google Drive.**



**Your Phone**



**Internet**



**Google Data  
Center**

*Your files are NOT inside your mobile.  
They live on Google's servers —  
download them anytime.*



# Real-Life Example 2 — Gmail



**You send an email to a friend. Where is it stored?**

Not on your laptop — it's stored on Google's cloud servers.

*Even if your laptop breaks, you can log in from any other computer — and your email is still there. Because it lives in the Cloud.*

*Key idea: your data outlives your device — because it's not stored on the device at all.*

Imagine you're starting a company. You need:

## Why Do We Need Cloud Computing?



**Servers**



**Storage**



**Database**



**Software**

### Buy Everything Yourself



₹20 Lakh or more — upfront hardware cost

### Rent From the Cloud



Pay only for what you use — scale up or down anytime

*This is exactly why startups and enterprises alike choose cloud computing.*

# Advantages of Cloud Computing



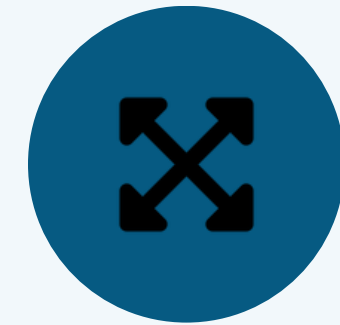
## Cost Saving

No need to buy expensive hardware.  
A startup rents servers instead of buying them.



## Access Anywhere

Access your files from home, college, office, or mobile — as long as you have Internet.



## Large Storage

Need more space?  
Just upgrade your cloud plan — no need to buy a new hard disk.



# Advantages of Cloud Computing



## Automatic Backup

Cloud providers automatically back up your data — reducing the risk of loss.



## High Security

Large cloud companies invest heavily in protecting your data with strong safeguards.



## Easy Collaboration

Multiple people can work on the same document at once — like Google Docs.

# Disadvantages of Cloud Computing



Internet connection is required to access your data



Data privacy concerns — trusting a third party with your data



Ongoing monthly subscription costs



Service outages can temporarily affect access

# Popular Cloud Services We Use Daily

|                                                                                    |                           |                |                                                                                      |                          |                        |
|------------------------------------------------------------------------------------|---------------------------|----------------|--------------------------------------------------------------------------------------|--------------------------|------------------------|
|   | <b>Gmail</b>              | <i>Email</i>   |   | <b>YouTube / Netflix</b> | <i>Video Streaming</i> |
|   | <b>Google Drive</b>       | <i>Storage</i> |   | <b>Spotify</b>           | <i>Music Streaming</i> |
|  | <b>Dropbox / OneDrive</b> | <i>Storage</i> |  | <b>Google Photos</b>     | <i>Photo Backup</i>    |

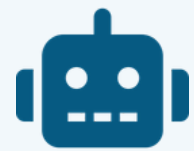


# Recent Trends in Cloud Computing

*Top 10 Cloud Computing Trends — as cloud adoption grows worldwide, providers like Google, Microsoft, IBM, and Amazon keep innovating to deliver faster, safer, and smarter services.*

# Top Cloud Trends (1–5)

1



## AI & Machine Learning

Self-learning, personalized cloud, and stronger data privacy powered by AI/ML.

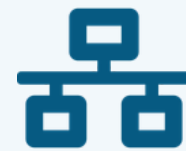
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## Data Security

Encryption, authentication, and recovery systems minimize breaches and data loss.

3



## Multi & Hybrid Cloud

Businesses split workloads across providers to cut risk and boost cost-efficiency.

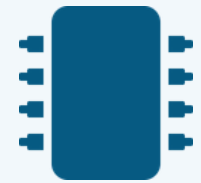
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## Low-Code / No-Code

Build apps and websites fast — no deep coding knowledge required (e.g. Figma, Zoho).

5



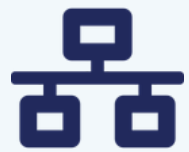
## Edge Computing

Processing data closer to its source for lower latency, aided by the rise of 5G.



# Top Cloud Trends (6–10)

6



## IoT

Sensors and devices generate huge data, stored and analyzed via cloud servers.

7



## Kubernetes & Docker

Automated scaling, deployment, and container management for large-scale apps.

8



## Serverless Computing

Pay-as-you-go execution — no server management, automatic scaling on demand.

9



## DevSecOps

Security built into the development pipeline to guard against cloud-based attacks.

10



## Disaster Recovery

Cloud backups protect critical data from outages, cyberattacks, and hardware failure.

*Also emerging: Blockchain, Cloud Gaming, and Quantum Computing.*